Growth Rates

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> **Remark**
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> *Remark.* The running time of an algorithm is a function of the size n of its input. To compare running times, we ignore fixed factors and small inputs. This page defines the notation for that comparison.

Upper bounds

Definition 0.1 (Big-O)**.** Let f and g be functions from the natural numbers to the real numbers. Then $f(n)$ is **big-O** of $g(n)$ [1], written $f(n) = O(g(n))$, when there exist a real number $c > 0$ and a natural number N such that, for every natural number $n \geq N$,

$$|f(n)| \leq c\,|g(n)|.$$

In words, from some point N on, the absolute value of $f(n)$ is at most a fixed multiple c of the absolute value of $g(n)$. The "there exist" is an existential quantifier, and the "for every" is a universal quantifier.

Example 0.1 (The function $3n + 5$ is big-O of n)**.** Let $f(n) = 3n + 5$ and $g(n) = n$. To show $f(n) = O(g(n))$ by Definition 0.1, choose $c = 4$ and $N = 5$. For every natural number $n \geq 5$,

$$\begin{aligned} |f(n)| &= 3n + 5 && (3n + 5 > 0) \\ &\leq 3n + n && (5 \leq n, \text{ since } n \geq 5) \\ &= 4n && (3n + n = 4n) \\ &= 4\,|g(n)|, && (n = |n|, \text{ since } n \geq 0) \end{aligned}$$

which is the inequality in Definition 0.1 with $c = 4$. Hence $3n + 5 = O(n)$.

Lower bounds

Definition 0.2 (Big-Omega)**.** Let f and g be functions from the natural numbers to the real numbers. Then $f(n)$ is **big-Omega** of $g(n)$ [1], written $f(n) = \Omega(g(n))$, when there exist a real number $c > 0$ and a natural number N such that, for every natural number $n \geq N$,

$$|f(n)| \geq c\,|g(n)|.$$

In words, from some point N on, the absolute value of $f(n)$ is at least a fixed positive multiple c of the absolute value of $g(n)$. Dividing by c shows that $f(n) = \Omega(g(n))$ exactly when $g(n) = O(f(n))$ in the sense of Definition 0.1.

Example 0.2 (The function $3n + 5$ is big-Omega of n)**.** Let $f(n) = 3n + 5$ and $g(n) = n$. To show $f(n) = \Omega(g(n))$ by Definition 0.2, choose $c = 3$ and $N = 1$. For every natural number $n \geq 1$,

$$\begin{aligned} |f(n)| &= 3n + 5 && (3n + 5 > 0) \\ &\geq 3n && (5 \geq 0) \\ &= 3\,|g(n)|, && (n = |n|, \text{ since } n \geq 0) \end{aligned}$$

which is the inequality in Definition 0.2 with $c = 3$. Hence $3n + 5 = \Omega(n)$.

Matching bounds

Definition 0.3 (Big-Theta)**.** Let f and g be functions from the natural numbers to the real numbers. Then $f(n)$ is **big-Theta** of $g(n)$ [1], written $f(n) = \Theta(g(n))$, when both $f(n) = O(g(n))$ (big-O) and $f(n) = \Omega(g(n))$ (big-Omega). In words, $f(n)$ and $g(n)$ are within fixed multiples of each other, from some point on.

Example 0.3 (The function $3n + 5$ is big-Theta of n)**.** Let $f(n) = 3n + 5$ and $g(n) = n$. By Example 0.1, $f(n) = O(n)$, and by Example 0.2, $f(n) = \Omega(n)$. Therefore $f(n) = \Theta(n)$ by Definition 0.3.

References

[1] Wikipedia contributors, "Big O notation — Wikipedia, the free encyclopedia." 2026. Available: https://en.wikipedia.org/wiki/Big_O_notation